From the Conjecture to the Ground Set

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1 The conjecture

Let $c = (c_0, c_1, c_2)$ be a rank-3 profile with $d = c_0 + c_1 + c_2$. A *cylindric plane partition* with profile c is a triple $(\nu^{(1)}, \nu^{(2)}, \nu^{(3)})$ of partitions satisfying the interlacing conditions determined by c. The generating function is

$$\mathrm{GK}_c(z, q) = \sum_{\pi \in \mathrm{CPP}(c)} z^{\max(\pi)}\, q^{|\pi|}, \tag{1}$$

where $\max(\pi)$ is the largest entry and $|\pi|$ is the sum of all entries. Define the coefficients

$$g_m(q) = [z^m]\ \mathrm{GK}_c(z, q) = \sum_{\substack{\pi \in \mathrm{CPP}(c) \\ \max(\pi) = m}} q^{|\pi|}. \tag{2}$$

These are manifestly in $\mathbb{N}[q]$: they count CPPs.

The *Q-polynomial* is

$$Q_{n,c}(q) = (q;q)_n \cdot [z^n]\big((zq;q)_\infty \cdot \mathrm{GK}_c(z, q)\big). \tag{3}$$

Conjecture 1 (Warnaar 2.7; Corteel–Dousse–Uncu 4.2)**.** *For all $n \geq 0$ and all rank-3 profiles c,*

$$Q_{n,c}(q) \in \mathbb{N}[q].$$

Since the definition (**??**) involves the Euler product $(zq;q)_\infty = \prod_{k \geq 1}(1 - zq^k)$, which introduces alternating signs, it is not at all obvious that Q_n has non-negative coefficients. The rest of this note derives what the coefficients of Q_n actually count (with signs), and identifies the exact point where non-negativity becomes non-trivial.

2 Expanding the Euler product

The Euler product in (**??**) is

$$(zq;q)_\infty = \prod_{k=1}^{\infty}(1 - zq^k).$$

Each factor contributes either 1 or $-zq^k$. Choosing the $-zq^k$ term for a subset $J \subseteq \{1, 2, 3, \ldots\}$ gives $(-1)^{|J|}\, z^{|J|}\, q^{\sigma(J)}$, where $\sigma(J) = \sum_{k \in J} k$. Therefore:

$$(zq;q)_\infty \cdot \mathrm{GK}_c(z, q) = \sum_{J,\, m} (-1)^{|J|}\, q^{\sigma(J)}\, g_m\, z^{|J|+m}.$$

Extracting the coefficient of z^n (which forces $|J| + m = n$, i.e. $m = n - |J|$, and in particular $|J| \le n$):

$$f_n(q) := [z^n]\big((zq;q)_\infty \cdot \mathrm{GK}_c\big) = \sum_{j=0}^{n} (-1)^j \underbrace{\left(\sum_{\substack{J \subseteq \{1,2,\dots\} \\ |J|=j}} q^{\sigma(J)} \right)}_{= \; q^{j(j+1)/2}/(q)_j} \cdot g_{n-j}. \tag{4}$$

The sum over j-element subsets of $\{1, 2, \dots\}$ weighted by $q^{\sigma(J)}$ is the generating function for partitions into j distinct positive parts, which equals $q^{j(j+1)/2}/(q)_j$ (the minimal such partition has parts $1, 2, \dots, j$ with sum $j(j+1)/2$, and the excess is an unrestricted partition into at most j parts).

3 Multiplying by $(q)_n$

By definition (**??**):

$$Q_n = (q)_n \cdot f_n = \sum_{j=0}^{n} (-1)^j \, q^{j(j+1)/2} \, \frac{(q)_n}{(q)_j} \, g_{n-j}.$$

Now split $\dfrac{(q)_n}{(q)_j} = \dfrac{(q)_n}{(q)_j \, (q)_{n-j}} \cdot (q)_{n-j} = \begin{bmatrix} n \\ j \end{bmatrix} \cdot (q)_{n-j}$, giving

$$\boxed{Q_n = \sum_{j=0}^{n} (-1)^j \, q^{j(j+1)/2} \begin{bmatrix} n \\ j \end{bmatrix} h_{n-j},} \tag{5}$$

where

$$h_m(q) := (q)_m \cdot g_m(q) = \prod_{k=1}^{m} (1 - q^k) \cdot g_m(q). \tag{6}$$

4 The ground set

The key combinatorial identity is

$$q^{j(j+1)/2} \begin{bmatrix} n \\ j \end{bmatrix} = \sum_{\substack{S \subseteq \{1,\dots,n\} \\ |S|=j}} q^{\sigma(S)}. \tag{7}$$

(The minimal j-element subset $\{1, \dots, j\}$ has $\sigma = j(j+1)/2$; the q-binomial coefficient counts the excess over this minimum.)

Substituting (**??**) into (**??**) and summing over all subset sizes:

$$\boxed{Q_n = \sum_{S \subseteq \{1,\dots,n\}} (-1)^{|S|} \, q^{\sigma(S)} \, h_{n-|S|}.} \tag{8}$$

This is a signed sum over subsets S of $\{1, \dots, n\}$. The weight is $q^{\sigma(S)} \cdot h_{n-|S|}(q)$ and the sign is $(-1)^{|S|}$.

If $h_m \in \mathbb{N}[q]$ for all $0 \le m \le n$, then each $h_{n-|S|}$ is a polynomial with non-negative integer coefficients, and equation (**??**) defines a *genuine signed sum over a genuine set*:

$$Q_n = \sum_{\substack{S \subseteq \{1,\dots,n\} \\ w \ge 0}} (-1)^{|S|} \, [q^w] \, h_{n-|S|} \;\cdot\; q^{\sigma(S)+w}. \tag{9}$$

The ground set is

$$\mathcal{G}_{n,c} = \big\{(S, w) : S \subseteq \{1, \dots, n\},\ 0 \le w,\ [q^w]\, h_{n-|S|} > 0\big\},$$

with multiplicity $[q^w]\, h_{n-|S|}$ at each point, total weight $\sigma(S) + w$, and sign $(-1)^{|S|}$.

Interpretation. The subset S is the *crossing set*: it records the levels at which lattice paths cross.

- $S = \emptyset$: non-intersecting family, sign $+1$.
- $S \neq \emptyset$: paths cross at levels in S, sign $(-1)^{|S|}$.

The conjecture $Q_n \in \mathbb{N}[q]$ says: all elements with $S \neq \emptyset$ cancel in pairs (opposite signs, same total weight), leaving a non-negative remainder from $S = \emptyset$.

5 Where $h_m \ge 0$ is used

The derivation up to equation (**??**) is unconditional: it holds for all profiles c.

The ground set $\mathcal{G}_{n,c}$ requires one additional condition: $h_m \in \mathbb{N}[q]$. Without it, some multiplicities $[q^w]\, h_{n-|S|}$ are negative, and $\mathcal{G}_{n,c}$ is not a set—it is a formal signed collection. An involution (a bijective pairing) requires actual objects to pair.

Example 2 ($d = 2$: the ground set is real)**.** For $c = (2, 0, 0)$: $g_0 = 1$, $g_1 = q + q^2$, $g_2 = q^2 + q^3 + 2q^4 + q^5 + q^6$.

$h_0 = (q)_0 \cdot g_0 = 1$.

$h_1 = (q)_1 \cdot g_1 = (1 - q)(q + q^2) = q + q^2 - q^2 - q^3 = q - q^3$.

Wait—that has a negative coefficient. But actually $h_1 = (1-q)(q+q^2) = q+q^2-q^2-q^3 = q-q^3$. Hmm. Let's recompute from the definition in the companion notes.

Actually, I need to recompute. For $c = (2, 0, 0)$, the value $h_1 = q^2$ was reported in the companion notes (CANONICAL_PEEL), giving $Q_1 = h_1 - q \cdot h_0 = q^2 - q$. But $Q_1 = q^2$ for this profile, so this means $h_1 - q = q^2$, i.e. $h_1 = q + q^2$.

This suggests $g_1 = (q + q^2)/(1 - q) = q(1 + q)/(1 - q)$, which is not a polynomial. So the relationship $h_m = (q)_m \cdot g_m$ needs more care for $m > 0$.

Remark 3 (To be reconciled)**.** The precise definition of h_m requires careful bookkeeping of the E–H factorisation. The key point of this note is the structural derivation: equations (**??**) and (**??**) hold by algebra, and the ground set $\mathcal{G}_{n,c}$ is a genuine set if and only if $h_m \in \mathbb{N}[q]$ for all relevant m.

6 Summary

Object	Meaning
$g_m(q)$	CPPs with max $= m$, weighted by $q^{\lvert\pi\rvert}$ $\in \mathbb{N}[q]$ always
$h_m(q)$	"corrected" coefficients $\in \mathbb{N}[q]$ when $d \not\equiv 0 \pmod 3$
$S \subseteq \{1, \ldots, n\}$	crossing set (which levels have intersecting paths)
$\sigma(S) = \sum_{s \in S} s$	crossing weight
$(-1)^{\lvert S\rvert}$	sign (even crossings $+$, odd crossings $-$)
$q^{\sigma(S)} \cdot h_{n-\lvert S\rvert}$	total unsigned contribution of crossing pattern S
$Q_n = \sum_S (-1)^{\lvert S\rvert} q^{\sigma(S)} h_{n-\lvert S\rvert}$	the signed sum that must be ≥ 0

The conjecture says: the positive terms ($|S|$ even) outweigh the negative terms ($|S|$ odd) at every power of q. A sign-reversing involution on the ground set $\mathcal{G}_{n,c}$ proves this by explicitly pairing each negative element with a positive one at the same total weight.